

Boolean functions to their representation -

A function of 'n' Boolean variables denoted by $f(x_1, x_2, \dots, x_n)$ takes one of the two possible values, 0 or 1. The various ways of representing a given function are as follows:-

1. Sum-of-products (SOP) form - (or Disjunctive Normal form (DNF))

ex:- $f(A, B, C) = \bar{A}B + \bar{B}C$

2. Product-of-sums (POS) form or Conjunctive Normal form (CNF) -

ex:- $f(A, B, C) = (\bar{A} + \bar{B})(B + C)$

3. Truth Table form - Here, function is specified by listing all possible combinations of values assumed by the variables and the corresponding values of the function.

ex: Truth table for $f(A, B, C) = \bar{A}B + \bar{B}C$

A	B	C	$f(A, B, C)$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

✓ Standard sum-of-products form -

(or Disjunctive canonical form, DCF or Expanded SOP form or)
Canonical SOP form

In this form, the function is the sum of a number of product terms where each product term contains all the variables of the function in either complemented or uncomplemented form.

ex:- Deriving from truth table - By finding the sum of all the terms that correspond to those combinations (rows) for which 'f' assumes the value 1.
 from above truth table, $f = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}C + A\bar{B}\bar{C}$
 (uncomplemented if value = 1, complemented if value = 0)

Deriving from SOP algebraically -

$$\begin{aligned} f(A, B, C) &= \bar{A}B + \bar{B}C \\ &= \bar{A}B(1+C) + \bar{B}C(A+\bar{A}) \\ &= \bar{A}BC + \bar{A}B\bar{C} + \bar{A}\bar{B}C + A\bar{B}C \end{aligned}$$

Multiply with $(x+\bar{x})$ in product term for each variable x that is missing

A product term which contains all the variables of the function either in complemented or uncomplemented form is called a minterm.

An n variable function can have 2^n minterms.

Standard SOP is equal to sum of minterms whose value is equal to 1.

Minterms are denoted as $m_0, m_1, m_2, \dots$ where suffixes are decimal codes.

$$\therefore f(A, B, C) = m_1 + m_2 + m_3 + m_5$$

$$\text{Also, } f(A, B, C) = \sum m(1, 2, 3, 5)$$

sum of all minterms with decimal codes given in the parenthesis.

5. Standard Product-of-Sums form -

(conjunctive canonical form, CCF or Expanded Product-of-Sums form or Canonical POS form)

This is derived by considering the combinations for which $f=0$. Each term is the sum of all variables. A variable appears in uncomplemented form if value = 0 or appears in the complemented form if value = 1.

ex:- Deriving from truth table - find out the rows for which 'f' assumes 0.

$$f(A, B, C) = (A+B+C) (\bar{A}+B+C) * (\bar{A}+\bar{B}+C) (\bar{A}+\bar{B}+\bar{C})$$

uncomplemented if value = 0
complemented if value = 1

Deriving from POS algebraically -

$$\begin{aligned} f(A, B, C) &= (\bar{A}+\bar{B})(B+C) \\ &= (\bar{A}+\bar{B}+C\bar{C})(A\bar{A}+B+C) \\ &= (\bar{A}+\bar{B}+C)(\bar{A}+\bar{B}+\bar{C})(A+B+C)(\bar{A}+B+C) \end{aligned}$$

Add $x\bar{x}$ to the sum term for each variable x that is missing.

Maxterms

A sum term which contains each of the n variables in either complemented or uncomplemented form is called a maxterm.

for n variable function, there will be atmost 2^n maxterms

Standard or canonical POS = Product of maxterms for which $f=0$.

Also, $f(A, B, C) = M_0 * M_4 * M_6 * M_7$

Also, $f(A, B, C) = \prod M(0, 4, 6, 7)$

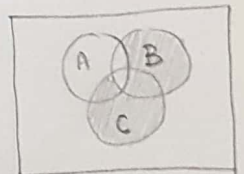
Maxterms are denoted as $M_0, M_1, M_2, \dots$ where suffixes are decimal codes.

product of all maxterms with decimal codes given in the parenthesis.

We can also use, ' Λ ' in place of ' Π '.

Other ways of representing a Boolean function are -

6. Venn diagram form - $f(A, B, C) = \bar{A}B + \bar{B}C$ is represented as:-



7. Octal designation - for $m_7, m_6, m_5, m_4, m_3, m_2, m_1, m_0$

0 0 1 0 1 1 1 0

we write $f(A, B, C) = (0, 5, 6)_8$

8. Karnaugh map -

Exercise -

Ques. 1. Convert into standard SOP

i) $Y(A, B) = A + B$

ii) $Y(A, B, C) = A + BC$

$$\begin{aligned} \text{i) } Y(A, B) &= A + B \\ &= A(B + \bar{B}) + B(A + \bar{A}) \\ &= AB + A\bar{B} + AB + \bar{A}B \\ &= AB + \bar{A}B + A\bar{B} \end{aligned}$$

$$\begin{aligned} \text{ii) } Y(A, B, C) &= A + BC \\ &= A(B + \bar{B})(C + \bar{C}) + BC(A + \bar{A}) \\ &= (AB + A\bar{B})(C + \bar{C}) + ABC + \bar{A}BC \\ &= ABC + AB\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + ABC + \bar{A}BC \\ &= ABC + AB\bar{C} + A\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} \\ &\quad \begin{matrix} 111 & 110 & 101 & 011 & 100 \end{matrix} \\ &= \sum m(3, 4, 5, 6, 7) \end{aligned}$$

Ques. 2. Convert into standard POS

i) $Y(A, B, C) = (A + \bar{B})(B + C)(A + \bar{C})$

$$\begin{aligned} Y(A, B, C) &= (A + \bar{B} + C\bar{C})(B + C + A\bar{A})(A + \bar{C} + B\bar{B}) \\ &= (A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + B + C)(\bar{A} + B + C)(A + B + \bar{C})(A + \bar{B} + \bar{C}) \\ &= (A + B + C)(\bar{A} + B + C)(A + \bar{B} + C)(A + B + \bar{C})(A + \bar{B} + \bar{C}) \\ &\quad \begin{matrix} (000) & (100) & (010) & (001) & (011) \end{matrix} \\ &= \prod M(0, 1, 2, 3, 4) \end{aligned}$$

Conversion between Canonical forms -

To convert one canonical form to another, interchange the symbols Σ and Π , and list those numbers missing from the original form.

$$\text{ex:- } f(A, B, C) = \Sigma m(0, 2, 4, 6, 7) \checkmark$$

$$\overline{f(A, B, C)} = \Sigma m(1, 3, 5) = (m_1 + m_3 + m_5)$$

$$\begin{aligned} f(A, B, C) &= \overline{\overline{f(A, B, C)}} = \overline{m_1 + m_3 + m_5} \\ &= \overline{m_1} \cdot \overline{m_3} \cdot \overline{m_5} \\ &= M_1 \cdot M_3 \cdot M_5 \end{aligned}$$

$$\therefore \boxed{\Sigma m(0, 2, 4, 6, 7) = \Pi M(1, 3, 5)}$$
$$\boxed{\overline{m_j} = M_j} \checkmark$$

ex:- Expand $\overline{A+B}$ to minterms and maxterms.

$$\begin{aligned} f(A, B) &= \overline{A+B} \\ &= \overline{A(B+\overline{B}) + \overline{B}(A+\overline{A})} \\ &= \overline{AB + \overline{A}\overline{B} + A\overline{B} + \overline{A}B} \\ &= \overline{AB + A\overline{B} + \overline{A}\overline{B}} \\ &= \Sigma m(0, 1, 2). \end{aligned}$$

$$\therefore \text{POS} = \Pi M(3) = \overline{A+B}$$

Ques. Expand $A + B\overline{C} + AB\overline{D} + ABCD$ to minterms and maxterms.

$$\begin{aligned} A &\rightarrow A(B+\overline{B})(C+\overline{C})(D+\overline{D}) \\ &= (AB + A\overline{B})(C+\overline{C})(D+\overline{D}) \\ &= (AB\overline{C} + A\overline{B}\overline{C} + ABC + A\overline{B}C)(D+\overline{D}) \\ &= (AB\overline{C}D + A\overline{B}\overline{C}D + ABC\overline{D} + A\overline{B}CD + AB\overline{C}\overline{D} + A\overline{B}C\overline{D} + ABCD + A\overline{B}C\overline{D}) \end{aligned}$$

$$\begin{aligned} B\overline{C} &\rightarrow B\overline{C}(A+\overline{A})(D+\overline{D}) \\ &= (AB\overline{C} + \overline{A}B\overline{C})(D+\overline{D}) \\ &= AB\overline{C}D + \overline{A}B\overline{C}D + AB\overline{C}\overline{D} + \overline{A}B\overline{C}\overline{D} \end{aligned}$$

$$\begin{aligned} AB\overline{D} &\rightarrow AB\overline{D}(C+\overline{C}) \\ &= AB\overline{D}C + AB\overline{D}\overline{C} \end{aligned}$$

$$ABCD \rightarrow ABCD$$

$$\begin{aligned} \therefore f(A, B, C, D) &= \underbrace{ABCD}_{m_5} + \underbrace{AB\bar{C}D}_{m_3} + \underbrace{A\bar{B}CD}_{m_{11}} + \underbrace{A\bar{B}\bar{C}D}_{m_9} + \underbrace{ABC\bar{D}}_{m_{14}} + \underbrace{AB\bar{C}\bar{D}}_{m_{12}} \\ &\quad + \underbrace{A\bar{B}C\bar{D}}_{m_{10}} + \underbrace{A\bar{B}\bar{C}\bar{D}}_{m_8} + \underbrace{\bar{A}BCD}_{m_7} + \underbrace{\bar{A}B\bar{C}D}_{m_6} \\ &= \sum m(4, 5, 8, 9, 10, 11, 12, 13, 14, 15) \end{aligned}$$

$$\therefore \text{POS} = \prod M(0, 1, 2, 3, 6, 7)$$

Minimization of Switching functions

Boolean expressions can be simplified algebraically but being not a systematic method we can never be sure whether minimal expression obtained is real minimal or not. It depends on our familiarity with the laws & ability to apply them.

The Karnaugh map (K-map) method is a systematic method of simplifying Boolean expressions.

The k-map is a chart or a graph, composed of an arrangement of adjacent cells, each representing a particular combination of variables in sum or product form. Like a truth table, it shows the relationship between the logic inputs and the desired outputs.

- * k-map is usually limited to 6 variables.
- * An n-variable function can have 2^n possible combinations of product terms (minterms) in SOP form or 2^n maxterms in POS form.
- * $\therefore$, K-map is a graphical representation of Boolean expressions, a two variable k-map will have $2^2 = 4$ cells or squares, a three " " " " $2^3 = 8$ cells or squares, a four " " " " $2^4 = 16$ cells.

Two - variable k-map -

$\therefore$, $2^2 = 4$ $\therefore$, it is possible to have 4 combinations of input variables A and B.

In SOP form, these are $\bar{A}\bar{B}$, $\bar{A}B$, $A\bar{B}$ and AB called as minterms

$\therefore$, 2-variable k-map has 4 squares or cells; each square represents a unique minterm as shown below:-

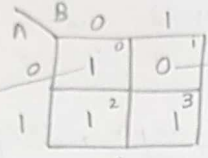
A \ B	0	1
0	$\bar{A}\bar{B}^0$	$\bar{A}B^1$
1	$A\bar{B}^2$	AB^3

The minterms of a 2-variable k-map

ex:- consider the expression $f = \bar{A}\bar{B} + \bar{A}B + AB$
 $= m_0 + m_2 + m_3$
 $f = \sum m(0, 2, 3)$

A	B	f
0	0	1
0	1	0
1	0	1
1	1	1

The mapping of above SOP expression is shown as follows:-

It indicates that corresponding minterm is included in the output expression.  This minterm does not appear in the o/p.

K-map of $\sum m(0, 2, 3)$

Minimization of SOP expressions -

Look for adjacent squares having 1s and combine them to form larger squares to eliminate some variables. Two squares are said to be adjacent to each other - if their minterms differ in only one variable.

Two minterms, which are adjacent to each other, can be combined to form a bigger square called a 2-square or a pair. This eliminates one variable - the variable that is not common to both the minterms. (or, keep the common variables)

↓ how?

m_0 & m_2 can be combined to yield

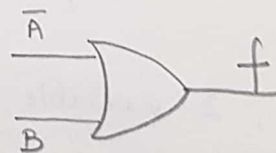
$$f_2 = m_0 + m_2 = \bar{A}\bar{B} + A\bar{B} = \bar{B}(A + \bar{A}) = \bar{B}.1 = \bar{B}$$

ex:- Reduce the expression $f = \bar{A}\bar{B} + \bar{A}B + AB$ using mapping.

$$f = m_0 + m_1 + m_3$$

$$= \sum m(0, 1, 3)$$

A \ B	0	1
0	1	1
1	0	1



$$f = \bar{A} + B$$

Mapping and minimization of POS expressions -

A function in two variables has 4 possible maxterms -
 (A, B) $A+B$, $A+\bar{B}$, $\bar{A}+B$, $\bar{A}+\bar{B}$
 M_0 M_1 M_2 M_3

B	0	1
A 0	$A+B$ 0	$A+\bar{B}$ 1
A 1	$\bar{A}+B$ 2	$\bar{A}+\bar{B}$ 3

The maxterms of a two variable K-map

ex:- Plot the expression $f = (A+B)(\bar{A}+B)(\bar{A}+\bar{B})$ on the K-map.
 $= \Pi M(0, 2, 3)$

B	0	1
A 0	0 ⁰	1 ¹
A 1	0 ²	0 ³

K-map for $\Pi M(0, 2, 3)$

Here, 0's are placed in the squares corresponding to the maxterms.

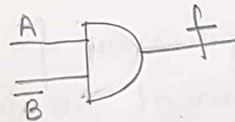
Minimization -

Combine the adjacent 0's into as large as possible. Read the common variables - if constant value is 1, put it as complemented variable & if constant value is 0, put it as non-complemented variable & then write them as a sum term.

ex:- Reduce the expression $f = (A+B)(A+\bar{B})(\bar{A}+\bar{B})$ using mapping.

B	0	1
A 0	0 ⁰	0 ¹
A 1	1 ²	0 ³

$f = A \cdot \bar{B}$



Three-Variable K-map -

A function in three variables (A, B, C) expressed in SOP can have 8 possible combinations: $\bar{A}\bar{B}\bar{C}$, $\bar{A}\bar{B}C$, $\bar{A}B\bar{C}$, $\bar{A}BC$, $A\bar{B}\bar{C}$, $A\bar{B}C$, $AB\bar{C}$, ABC .

In POS form, 8 possible combinations are:-

$A+B+C$, $A+B+\bar{C}$, $A+\bar{B}+C$, $A+\bar{B}+\bar{C}$, $\bar{A}+B+C$, $\bar{A}+B+\bar{C}$,
 $\bar{A}+\bar{B}+C$ and $\bar{A}+\bar{B}+\bar{C}$

A \ BC	00	01	11	10
0	$\bar{A}\bar{B}\bar{C}^0$	$\bar{A}B\bar{C}^1$	$A\bar{B}\bar{C}^3$	$\bar{A}B\bar{C}^2$
1	$A\bar{B}\bar{C}^4$	$A\bar{B}C^5$	ABC^7	$AB\bar{C}^6$

A \ BC	00	01	11	10
0	$A+B+C^0$	$A+B+\bar{C}^1$	$A+\bar{B}+\bar{C}^3$	$A+\bar{B}+C^2$
1	$\bar{A}+B+C^4$	$\bar{A}+B+\bar{C}^5$	$\bar{A}+\bar{B}+\bar{C}^7$	$\bar{A}+\bar{B}+C^6$

The three-variable k-map

Steps for minimization -

1. Plot the k-map & place 1s (0s) corresponding to the minterms (maxterms) of the SOP (POS) expression.
2. Check the k-map for 1s (0s) which are not adjacent to any other 1 (0). They are isolated minterms (maxterms).
3. Check for pairs, quads and octets, even if they contain some 1s (0s) which have already been combined.
4. Form the minimal expression by summing (multiplying) the product (sum) terms of all the groups.

ex:- Reduce the expression $f = \sum m(0, 2, 3, 4, 5, 6)$ using mapping & implement it in AOI logic as well as in NAND logic.

(Ans:- $f = \bar{C} + A\bar{B} + \bar{A}B$)

ex:- Reduce the expression $f = \prod M(0, 1, 2, 3, 4, 7)$ using mapping & implement it in AOI logic as well as in NOR logic.

(Ans:- $f = (B+C)(\bar{B}+\bar{C})(A)$)

four-variable k-map -

A four variable (A, B, C, D) expression can have $2^4 = 16$ possible combinations of i/p variables.

ex:- $\bar{A}\bar{B}\bar{C}\bar{D}, \bar{A}\bar{B}\bar{C}D, \dots, ABCD$ $m_0, m_1, \dots, m_{15}$
 $A+B+C+D, A+B+C+\bar{D}, \dots, \bar{A}+\bar{B}+\bar{C}+\bar{D}$ $M_0, M_1, \dots, M_{15}$

A 4-variable K-map has $2^4 = 16$ squares or cells & each cell on the map represents either a minterm or a maxterm.

CD	00	01	11	10
AB 00	0 $\bar{A}\bar{B}\bar{C}\bar{D}$	1 $\bar{A}\bar{B}\bar{C}D$	3 ...	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

The binary no. designations of the rows & columns are in the Gray code.

Here, 11 follows 01 & 10 follows 11.

This is called adjacency ordering.

Minterms of a 4-variable k-map

ex:- Reduce $f = \sum m(2, 3, 6, 7, 8, 10, 11, 13, 14)$.

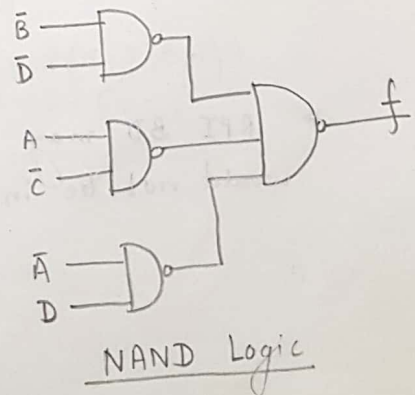
CD	00	01	11	10
AB 00	0	1	3 1	2 1
01	4	5	7 1	6 1
11	12	13 1	15	14 1
10	8 1	9	11 1	10 1

$$f = AB\bar{C}D + A\bar{B}\bar{D} + \bar{A}C + \bar{B}C + C\bar{D}$$

Ques. Reduce using mapping the expression $f = \sum m(0, 2, 3, 5, 7, 8, 9, 10, 12, 13)$ and implement the real minimal expression in universal logic.

Ans. $f_{min} = \bar{B}\bar{D} + A\bar{C} + \bar{A}D$

$$f_{min}(POS) = (A + \bar{B} + D)(\bar{A} + \bar{C} + \bar{D})(\bar{A} + \bar{B} + \bar{C})$$



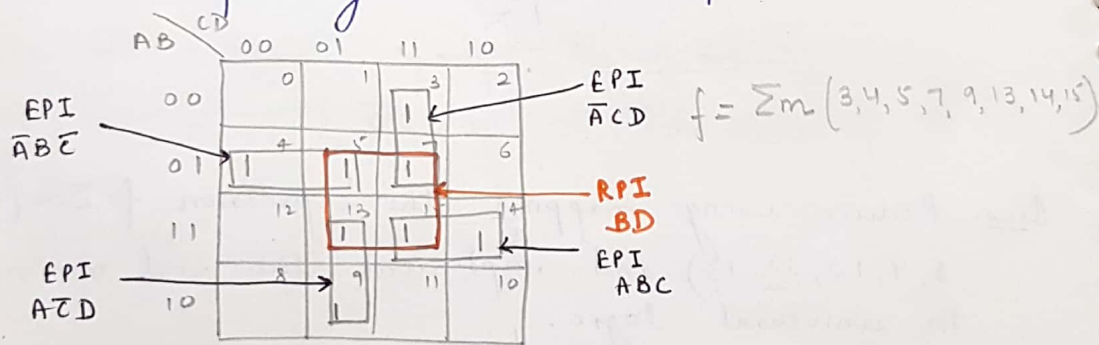
Ques. Reduce using mapping $f = \Pi M(2, 8, 9, 10, 11, 12, 14)$ & implement the real minimal expression in universal logic.

$$[f_{min} = (\bar{A}+B)(\bar{A}+D)(B+\bar{C}+D)]$$

Prime Implicants, Essential Prime Implicants, Redundant Prime Implicants & Selective Prime Implicants -

1. Each square or rectangle made up of the bunch of adjacent minterms is called a subcube. Each of these subcube is called a prime implicant (PI).
2. The prime implicant which contains atleast one 1 which cannot be covered by another PI is called an essential prime implicant (EPI).
3. The prime implicant whose each 1 is covered by atleast one EPI is called a redundant prime implicant (RPI).
4. A prime implicant which is neither an EPI nor a redundant prime implicant is called a selective prime implicant (SPI).

ex:- Consider the following 4-variable k-map:-



$$f(A, B, C, D) = \bar{A}\bar{C}D + ABC + \bar{A}\bar{C}D + \bar{A}\bar{B}\bar{C}$$

* RPI 'BD' may be included but the resulting expression would not be in minimal sum of products (MSP) form.

ex:- consider $F(A, B, C, D) = \sum m(0, 4, 5, 10, 11, 13, 15)$



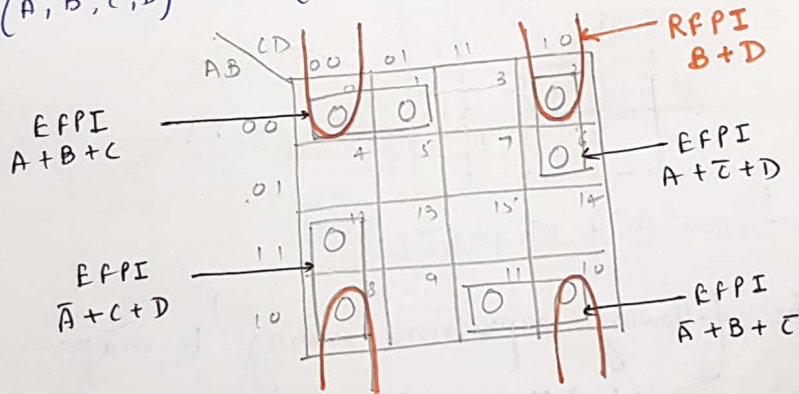
$$F(A, B, C, D) = \underbrace{\bar{A}\bar{C}\bar{D} + A\bar{B}\bar{C}}_{\text{EPIs}} + \underbrace{\bar{A}B\bar{C} + AB\bar{D}}_{\text{SPI}}$$

$$\text{or } F(A, B, C, D) = \underbrace{\bar{A}\bar{C}\bar{D} + A\bar{B}\bar{C}}_{\text{EPIs}} + \underbrace{\bar{B}\bar{C}\bar{D} + AB\bar{D}}_{\text{SPI}}$$

False Prime Implicants, Essential False Prime Implicants,
Redundant false Prime Implicants & selective false Prime
Implicants -

The maxterms are called false minterms. The prime implicants obtained by using the maxterms are called FPIs. The FPI which contains atleast one 0 which cannot be covered by any other FPI is called an essential false prime implicant (EFPI).

ex:- consider $F(A, B, C, D) = \sum m(3, 4, 5, 7, 9, 13, 14, 15) = \prod M(0, 1, 2, 6, 8, 10, 11, 12)$



Don't care combinations -

We have seen so far that for every combination of i/p, each min-term (or maxterm) has been specified as a 1 or 0.

It often occurs that for certain i/p combinations, value of the o/p is unspecified - either b'coz the i/p combinations are invalid or b'coz the precise value of the o/p is of no consequence.

- * The combinations for which the values of the expression are not specified are called don't care combinations (or optional combinations). The o/p is a don't care for these invalid combinations.

ex:- In excess-3 codes, binary states 0000, 0001, 0010, 1101, 1110 and 1111 are unspecified & never occur. These are called don't cares.

- * The don't care terms are denoted by - d, x or ϕ .

- * During the process of design of SOP, each don't care is treated as a 1 if it is helpful in map reduction; otherwise it is treated as a 0 & left alone.

ex:- Reduce the expression $f = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$.

AB \ CD	00	01	11	10
00	0	1	3	X
01	X	1	7	6
11	1	1	15	14
10	8	9	11	10

$$f_{\min} = B\bar{C} + B\bar{D} + \bar{A}\bar{C}D$$

Ques. Minimize the following expressions using k-map:-

a) $F(A, B, C, D) = \sum m(1, 4, 7, 10, 13) + \sum d(5, 14, 15)$

b) $f(A, B, C, D) = \sum m(1, 3, 7, 11, 15) + \sum d(0, 2, 5)$

[Ans. - $\bar{A}\bar{B}\bar{C} + \bar{A}\bar{C}D + BD + A\bar{C}\bar{D}$]
EPI

[Ans. - $\bar{A}\bar{B} + \bar{C}D$]
SPI EPI

Also show EPI & SPI on the k-map.